

Probabilistic model checking

Tomaž Jonatan Leonardis, Janez Ignacij Jereb, Sonja Šuković, Vanja Stojanović, Martin Žust

Motivation

We want to model a probabilistic system and check its properties.

Examples include:

- Randomised distributed algorithms
- Communication, network and multimedia protocols
- Security protocols
- Biological systems
- Planning and synthesis
- Performance and reliability
- Game theory
- Power management
- ...

How do you model a probabilistic system?

1. describe states and transitions
2. make number of state finite by abstraction
3. assign probabilities to transitions

Modeling customers on a shopping website

The screenshot displays the mimovrste website interface. At the top, navigation links include 'Sledenje naročila', 'Prezvezna mesta in dostava', 'Vračila in reklamacije', 'Kontakt', and 'Za podjetja'. A 'SMART TEDEN' banner is visible in the top right corner. The main header features the 'mimovrste' logo, a search bar with the placeholder 'Kaj iščete?', and links for 'Izberite oddelek', 'Vse akcije', 'Najdi', 'Prijava', and 'Koš'. A sidebar menu on the left lists various product categories with icons and arrows, including 'Računalništvo, telefonija', 'Gospodinski aparati', 'Audio-video in foto', 'Vrt in orodje', 'Šport in prosti čas', 'Igralne in otroške oprema', 'Avto-moto', 'Oprema doma', 'Za male živali', 'Drogerija', 'Kozmetika', 'Zdravje', 'Moda', 'Igre in razvedrilo', and 'Na voljo v trgovinah takoj'. The main content area features a large purple banner for 'SMART TEDEN' with a Persil Expert Sensitive gel bottle and an Apple MacBook Air 13, M4. The Persil bottle is priced at 15,99 € (reduced from 39,99 €). The MacBook Air is priced at 944,99 € (reduced from 1,049,99 €). The banner also includes the text '1 € = 1 leto brezplačne dostave' and 'Postanite SMART! >>'. Below the main banner, there are several smaller promotional tiles: 'VSE AKCIJE na enem mestu!', 'SMART 1 € = 1 leto brezplačne dostave', 'LEGO akcijska ponudba', 'Odpri embalaža DODATNIH - 10 % popusta', and 'Odpredaja zadnjih kosov'. At the bottom, there are two more banners: 'UŽIVANJE na vrtu' with a lawnmower and garden scene, and a Philips Head Pro shaver advertisement featuring a man shaving his head.

mimovrste Izberite oddelek **Vse akcije** **Najdi** **Prijava** **Koš**

SMART TEDEN

1 € = 1 leto
brezplačne dostave

Persil Expert Sensitive gel za pranje perila
39,99 € **-60%**
15,99 €

Apple MacBook Air 13, M4
1,049,99 € **-10%**
944,99 €

SMART TEDEN

Postanite SMART! >>

VSE AKCIJE na enem mestu!

SMART 1 € = 1 leto brezplačne dostave

LEGO akcijska ponudba

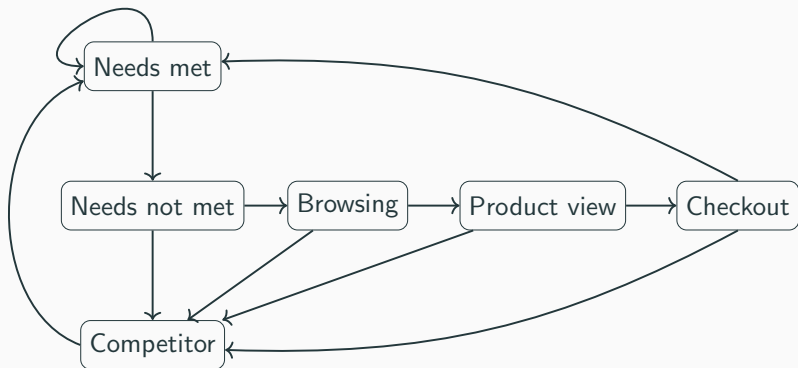
Odpri embalaža DODATNIH - 10 % popusta

Odpredaja zadnjih kosov

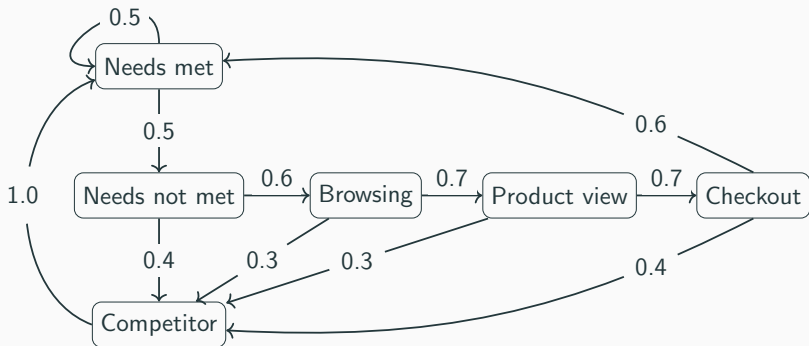
UŽIVANJE na vrtu
VRTHNE HIŠKE | POHIŠTVO | ŽARI | BAZENI
Prihranite >>

PHILIPS
Head Pro
Do konca.
Za drzne.
NOVO
VEČ >>

Modeling customers on a shopping website



Modeling customers on a shopping website



Formalize the model

- **Set of states**

$$S = \{\text{Needs_met}, \text{Needs_not_met}, \text{Competitor}, \dots\}$$

- **Initial state**

$$s_{\text{start}} = \text{Needs_met}$$

- **Transition probability matrix**

$$P : S \times S \rightarrow [0, 1]$$

$$\text{where } \sum_{s' \in S} P(s, s') = 1 \text{ for all } s \in S$$

Formalize the model

$$\begin{array}{l} \text{Needs_met} \\ \text{Needs_not_met} \\ \vdots \end{array} \begin{pmatrix} \text{Needs_met} & \text{Needs_not_met} \cdots \\ 0.5 & 0.5 & \cdots \\ 0 & 0 & \cdots \\ 1.0 & 0 & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

Additional features?

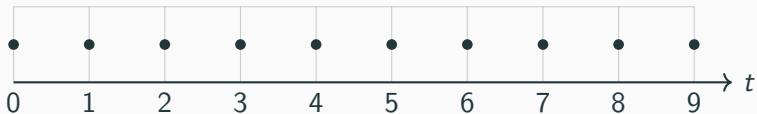
- Discrete vs continuous time
- Nondeterminism
- Rewards
- Intervals
- ...

The Many Kinds of Models

- Discrete-time Markov chains
- Continuous-time Markov chains
- Markov decision processes
- Markov automata
- Probabilistic timed automata
- ...

Continuous vs Discrete time

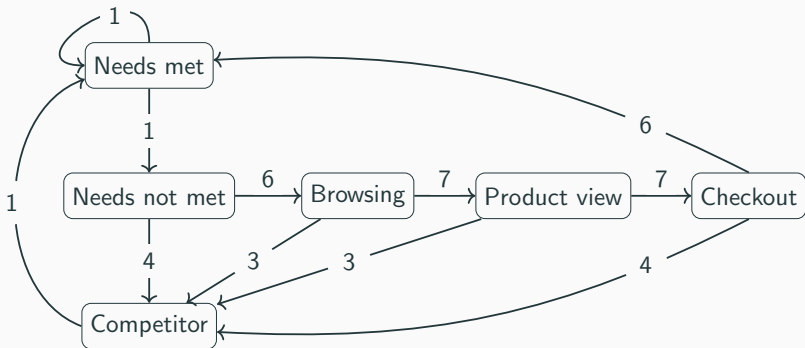
Discrete-Time (DTMC)



Continuous-Time (CTMC)



Continuous-time Markov chains



Continuous-time Markov chains

- **Set of states**

$$S = \{\text{Needs_met}, \text{Needs_not_met}, \text{Competitor}, \dots\}$$

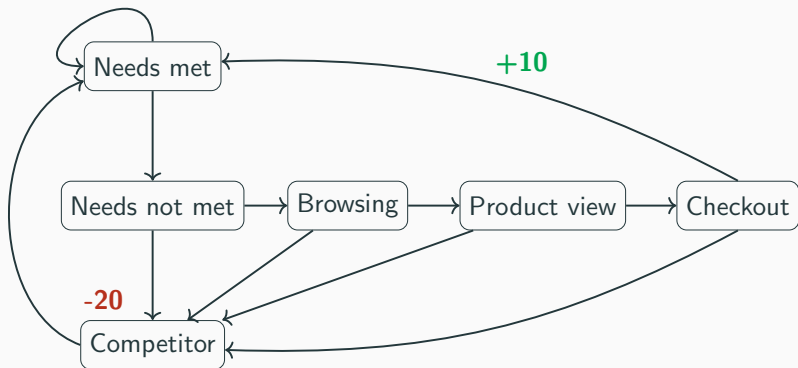
- **Initial state**

$$s_{\text{start}} = \text{Needs_met}$$

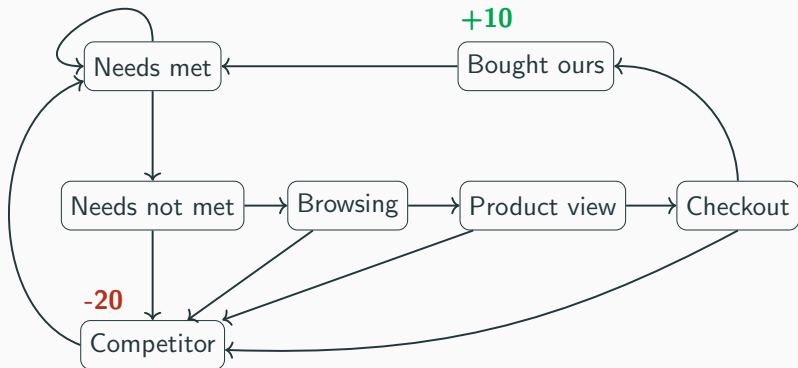
- **Transition rate matrix**

$$Q : S \times S \rightarrow [0, \infty)$$

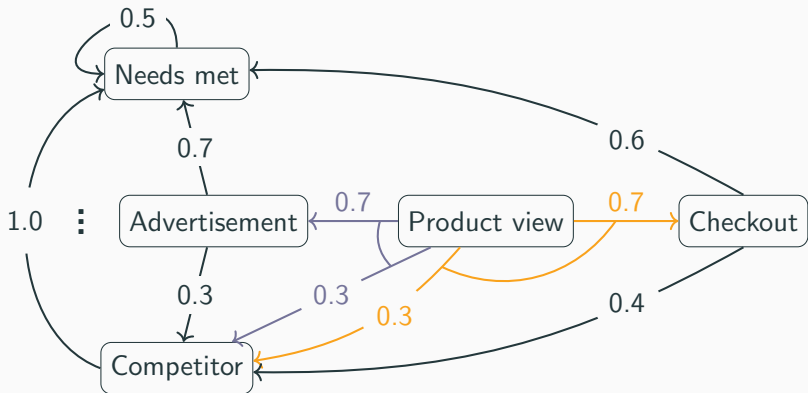
Rewards



Rewards



Nondeterminism



Formalize nondeterminism

- **Set of states**

$$S = \{\text{Needs_met}, \text{Advertisement}, \text{Competitor}, \dots\}$$

- **Initial state**

$$s_{\text{start}} = \text{Needs_met}$$

- **Transition probability function**

$$T : S \rightarrow 2^{\text{Dist}(S)}$$

$$T(\text{Product_view}) := \{\{\textit{Advertisement} \mapsto 0.7, \textit{Competitor} \mapsto 0.3\}, \\ \{\textit{Checkout} \mapsto 0.7, \textit{Competitor} \mapsto 0.3\}\}$$

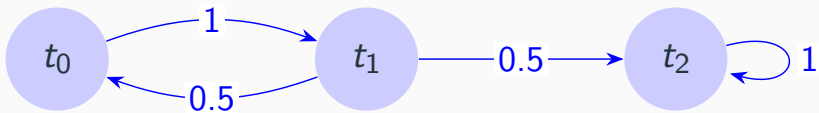
Table of types of models

	Discrete time	Continuous time
Deterministic	DTMC	CTMC
Nondeterministic	MDP	

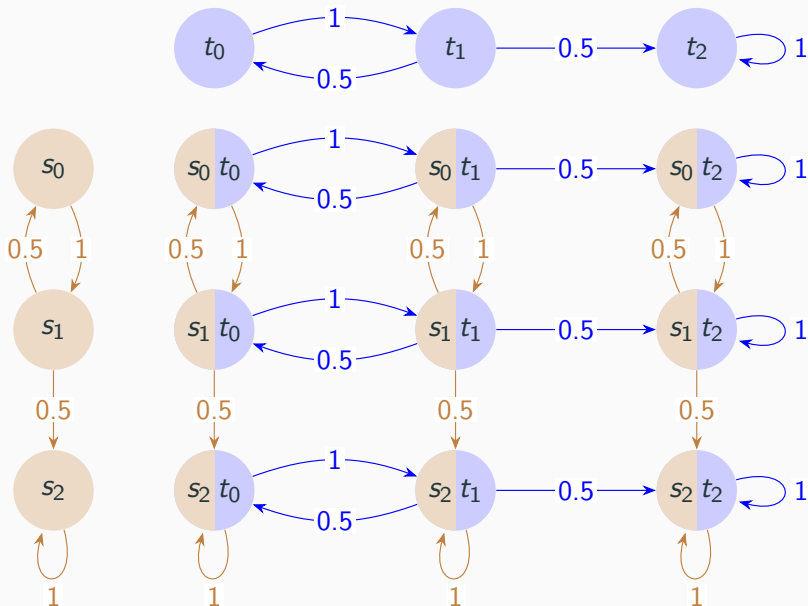
Other features include:

- uncertainty (IMDP, IDTMC)
- timing (PTA)
- partial observability (POMDP, POPTA)
- game-theoretic models (SMG, TSG, CSG, TPTG)

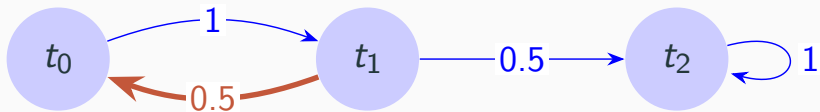
Simple Parallel Composition



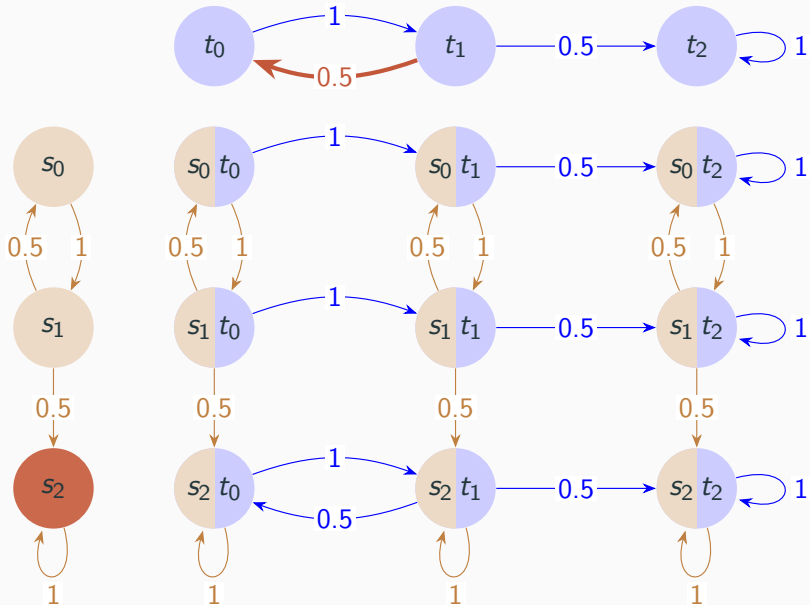
Simple Parallel Composition



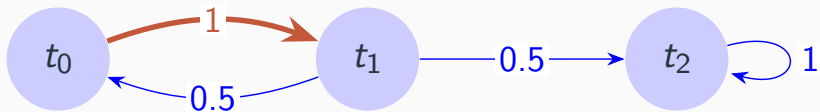
Simple Parallel Composition - node-edge synchronization



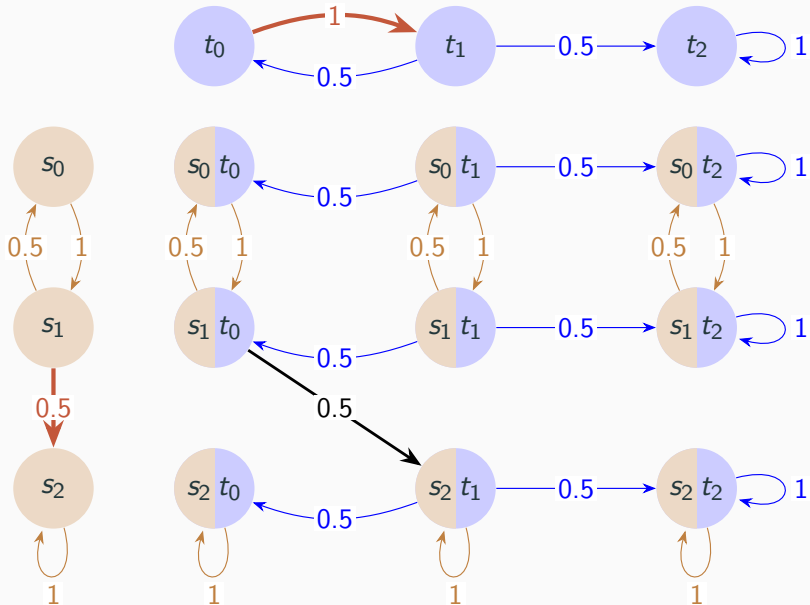
Simple Parallel Composition - node-edge synchronization



Simple Parallel Composition - synchronize on edges



Simple Parallel Composition - synchronize on edges



CTL - Computation Tree Logic

PCTL - **Probabilistic** Computation Tree Logic

addition: probabilistic operator P

CTL vs PCTL - definitions

- State formulae:

$$\phi ::= \text{true} \mid a \mid \phi \wedge \phi \mid \neg \phi \mid \forall \psi \mid \exists, \psi$$

$$\phi ::= \text{true} \mid a \mid \phi \wedge \phi \mid \neg \phi \mid P_{\sim p}[\psi],$$

where a is an atomic proposition, $p \in [0, 1]$ is a probability bound, $\sim \in \{<, >, \leq, \geq\}$, $k \in \mathbb{N}$

- Path formulae:

$$\psi ::= X \phi \mid F \phi \mid G \phi \mid \phi U \phi$$

where X means "next", G means "globally", and U means "until"

$$\psi ::= X \phi \mid \phi U^{\leq k} \phi \mid \phi U \phi,$$

where X means "next", U means "until", and $U^{\leq k}$ means "bounded until"

(path formulae only occur inside the P operator)

CTL vs PCTL - semantics of state formulae

$s \models \phi$ denotes “ s satisfies ϕ ” or “ ϕ is true in s ”

For a state s of an LTS $(S, s_{\text{init}}, \rightarrow, L)$:

$$s \models \text{true} \quad \text{always}$$

$$s \models a \iff a \in L(s)$$

$$s \models \phi_1 \wedge \phi_2 \iff s \models \phi_1 \text{ and } s \models \phi_2$$

$$s \models \neg\phi \iff s \not\models \phi$$

$$s \models \forall\psi \iff \omega \models \psi \text{ for all } \omega \in \text{Path}(s)$$

$$s \models \exists\psi \iff \omega \models \psi \text{ for some } \omega \in \text{Path}(s)$$

(we will mention operators $\forall$ and $\exists$ in PCTL later)

CTL vs PCTL - semantics of path formulae

$\omega \models \psi$ denotes “ ω satisfies ψ ” or “ ψ is true along ω ”

For a path ω of an LTS $(S, s_{\text{init}}, \rightarrow, L)$:

$$\omega \models X \phi \iff \omega(1) \models \phi$$

$$\omega \models F \phi \iff \exists k \geq 0 \text{ s.t. } \omega(k) \models \phi$$

$$\omega \models G \phi \iff \forall i \geq 0 \omega(i) \models \phi$$

$$\omega \models \phi_1 U \phi_2 \iff \exists k \geq 0 \text{ s.t. } \omega(k) \models \phi_2$$

$$\text{and } \forall i < k \omega(i) \models \phi_1$$

$$\omega \models \phi_1 U^{\leq k} \phi_2 \iff \exists i \leq k \text{ such that } \omega(i) \models \phi_2$$

$$\text{and } j < i, \omega(j) \models \phi_1$$

PCTL - semantics of the probability operator

Notation:

$$s \models P_{\sim p}[\psi]$$

means that the probability, from state s , that ψ is true for an outgoing path satisfies $\sim p$.

Example:

$$s \models P_{<0.15}[X \text{ succ}]$$

$\iff$ “the probability of succeed in the next state of outgoing paths from s is less than 0.15”

formally:

$$s \models P_{\sim p}[\psi] \iff \text{Prob}(s, \psi) \sim p$$

where:

$$\text{Prob}(s, \psi) = \Pr_s \{ \omega \in \text{Path}(s) \mid \omega \models \psi \}$$

PCTL - basic logical equivalences

False:

$$\text{false} \equiv \neg \text{true}$$

Disjunction:

$$\phi_1 \vee \phi_2 \equiv \neg(\neg\phi_1 \wedge \neg\phi_2)$$

Implication:

$$\phi_1 \rightarrow \phi_2 \equiv \neg\phi_1 \vee \phi_2$$

Negation example:

$$\neg P_{>p}[\phi_1 \ U \ \phi_2] \equiv P_{\leq p}[\phi_1 \ U \ \phi_2]$$

- "The probability that the sender has received n acknowledgements within k clock-ticks is at least 0.8":

$$P_{\geq 0.8} \left[F^{\leq k} \text{reply_count} = n \right]$$

- "The probability that component B fails before component A is less than 0.4"

$$P_{< 0.4} [\neg \text{fail}_A \ U \ \text{fail}_B]$$

We use P operator to define $\forall$ and $\exists$.

- **Qualitative** PCTL properties

$P_{\sim p}[\psi]$, where p is either 0 or 1.

- **Quantitative** PCTL properties

$P_{\sim p}[\psi]$, where p is in the range $(0, 1)$.

$\implies$

$P_{>0}[F \phi]$ is identical to $\exists F \phi$

there exists a finite path to a ϕ -state

$P_{\geq 1}[F \phi]$ is (similar to but) weaker than $\forall F \phi$

a ϕ -state is reached “almost surely”

PCTL is very useful for specifying **probabilistic reachability** properties

- e.g. “what is the probability of eventually reaching a failure/success state?”
- most PCTL properties are variations of:

$$\phi \ U \ \psi$$

meaning: “stay in ϕ states until reaching a ψ state”

Therefore, PCTL mainly reasons about:

- reaching target states
- staying within allowed states beforehand
- probabilities of these events
- possibly within bounded time

However, PCTL has limited expressivity

- difficult to express complex long-term behaviours
- e.g. “every request is eventually acknowledged”
- e.g. “event A happens infinitely often”

More expressive temporal logics include:

- PCTL*, which combines probabilities with richer temporal reasoning

PRISM model checker, primarily developed at the University of Oxford, is the most widely used tool in the field that supports various types of models, including:

- Discrete-Time Markov Chains (DTMCs)
- Markov Decision Processes (MDPs)
- Continuous-Time Markov Chains (CTMCs)
- Probabilistic Timed Automata (PTAs)

Its primary strength lies in its use of symbolic data structures, specifically Binary Decision Diagrams (BDDs) and Multi-Terminal Binary Decision Diagrams (MTBDDs). These structures allow PRISM to represent and analyze models with billions of states by exploiting the structural regularity of the system.

PRISM includes several analysis engines:

- Symbolic Engine: Uses MTBDDs for both representation and numerical solution.
- Explicit Engine: Uses sparse matrices and is often faster for models with less regularity.
- Hybrid Engine: Combines the memory efficiency of MTBDDs with the speed of sparse matrices.
- Simulator: Supports Statistical Model Checking (SMC), which uses Monte Carlo simulation to estimate probabilities when the state space is too large for exhaustive analysis.

PRISM-games extends its capabilities to stochastic multi-player games (systems with multiple agents)

Diverse Application Domains

The versatility of probabilistic model checking has led to its adoption across a wide range of application domains, including:

- Randomised distributed algorithms
- Communication protocols and networks
- Computer security
- Autonomous systems and robotics
- Safety-critical systems
- Biology and human behavior
- AI and self-adaptive systems
- Many more...

In the next few slides, we will briefly explore some of these application areas and the types of questions that probabilistic model checking can help answer. So many applications that we cannot cover them all, but we will give you a flavour of the diversity of applications and the types of questions that can be answered.

:2509.12968v1 [cs.LO] 16 Sep 2025

Probabilistic Model Checking: Applications and Trends

Marta Kwiatkowska¹, Gethin Norman^{1,2}, and David Parker¹

¹ Department of Computer Science, University of Oxford, Oxford, UK
{marta.kwiatkowska,david.parker}@cs.ox.ac.uk

² School of Computing Science, University of Glasgow, Glasgow, UK
gethin.norman@glasgow.ac.uk

Abstract. Probabilistic model checking is an approach to the formal modelling and analysis of stochastic systems. Over the past twenty five years, the number of different formalisms and techniques developed in this field has grown considerably, as has the range of problems to which it has been applied. In this paper, we identify the main application domains in which probabilistic model checking has proved valuable and discuss how these have evolved over time. We summarise the key strands of the underlying theory and technologies that have contributed to these advances, and highlight examples which illustrate the benefits that probabilistic model checking can bring. The aim is to inform potential users of these techniques and to guide future developments in the field.

1 Introduction

Probabilistic model checking [10] is a technique for the formal verification of stochastic systems. Properties to be verified are specified in temporal logic and then algorithmically checked against a model of the system. Some of the earliest work in the field dates from the 1980s [76,31], where algorithms were developed for computing the probability that linear temporal logic specifications are satisfied by sequential or concurrent probabilistic programs. The primary motivation

Probabilistic Model Checking: Applications and Trends

Kwiatkowska et al., 2025

<https://arxiv.org/abs/2509.12968>



www.prismmodelchecker.org

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Case Studies

- Randomised distributed algorithms
- Communication, network and multimedia protocols
- Security
- Biology
- Planning and synthesis
- Game-theory
- Performance and reliability
- Power management
- CTMC benchmarks
- Miscellaneous

PRISM Case Studies

PRISM has been used to analyse a wide range of case studies in many different application domains. A non-exhaustive list of these is given below. In many cases, we provide detailed information about the case study, including PRISM language source code and experimental results. For others, we just include links to the relevant publication.

We are always happy to include details of externally developed case studies. If you would like to contribute content about your work with PRISM, or you want us to add a pointer to a publication about your PRISM-related work, please [contact us](#).

If you are interested in PRISM models for the purposes of **benchmarking**, see also the [PRISM benchmark suite](#).

Randomised distributed algorithms

These case studies examine the correctness and performance of various *randomised distributed algorithms* taken from the literature.

- [Randomised self-stabilising algorithms](#) (Herman) (Israeli & Jalfon) (Beauquier et al.) [KNP12a]
- [Randomised two process wait-free test-and-set](#) (Tromp & Vitanyi)
- [Synchronous leader election protocol](#) (Itai & Rodeh)
- [Asynchronous leader election protocol](#) (Itai & Rodeh)
- [Randomised dining philosophers](#) (Lehmann & Rabin)
- [Randomised dining philosophers](#) (Lynch, Saia & Segala)
- [Dining cryptographers](#) (Chaum)
- [Randomised mutual exclusion](#) (Rabin)
- [Randomised mutual exclusion](#) (Pruett & Zuck)
- [Randomised consensus protocol](#) (Aspnes & Herlihy) (with Cadence SMV and PRISM) [KNS01a]
- [Randomised consensus shared coin protocol](#) (Aspnes & Herlihy) (with PRISM) [KNS01a]
- [Byzantine agreement protocol](#) (Cachin, Kursawe & Shoup) (with Cadence SMV and PRISM) [KN02]
- [Byzantine agreement protocol](#) (Cachin, Kursawe & Shoup) (with PRISM) [KN02]
- [Rabin's Choice Coordination Problem](#) [NM10]
- (contributed by Utschukwu Ndubaku and Annabelle McIver)
- [Dice programs](#) (Knuth & Yao)

Other related case studies:

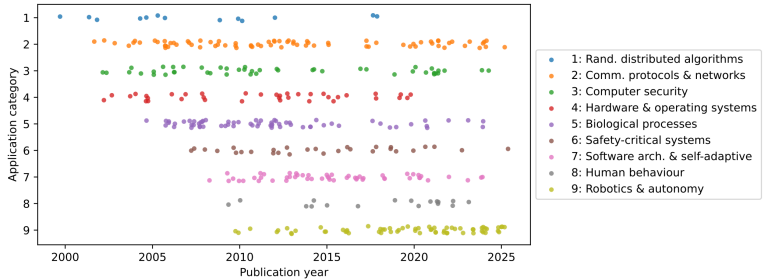
- [Probabilistic leader election algorithms](#) [FP04]
(by Wan Fokkink and Jun Pang)

Communication, network and multimedia protocols

PRISM Case Studies <https://www.prismmodelchecker.org/casestudies/index.php>

Recommended for further reading if you are interested in more details and references to specific papers in each area.

Categories of Application-Oriented Publications



Source: Kwiatkowska et al., 2025

Randomised Distributed Algorithms

- A fruitful initial area of study for probabilistic model checking

Problem: Symmetry breaking between concurrent processes - establishing correct termination and fairness

- Leader election
- Byzantine agreement
- Self-stabilization
- Consensus protocols
- Fault tolerance

Generally uses MDPs since they are ideally suited to model the mixture of stochasticity and concurrency, sometimes DTMCs are also used.

Scalability of verification techniques was enhanced with the development of symbolic methods.

Communication Protocols

- A natural progression from work on distributed systems
- Often deploy randomness to break symmetry
- In wireless we also model underlying medium's unreliability (message delays or losses)

Important examples:

- Bluetooth device discovery (multiple DTMC models with $\geq 10^{10}$ states)
- CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)
- FireWire and Zigbee protocols (use of PTAs)

Modern examples include wireless protocols, vehicular networks and wireless sensor networks

Key ideas:

- Attackers are unpredictable (randomized or adaptive attacks)
- Use of randomisation is widespread in security protocols (e.g. key generation)
- Need for strong guarantees (e.g. probability of attack success must be very low)

Natural use of MDPs to model the interaction between the system and an adversary. $\Rightarrow$ **Generalized by stochastic games**

Trade-off between security and performance (e.g. more randomisation can increase security but also increase latency) $\Rightarrow$ The expressivity of probabilistic temporal logics allows us to capture and analyze these trade-offs (e.g. relation between error and transmission cost).

CTMCs are useful for modeling stochastic dynamics of biological processes:

- Cellular processes (protein to protein interactions, gene expression, chemical reactions)
- Population dynamics (epidemic disease spreading, population growth)

Questions like: *“what is the expected number of molecules of protein X after 5 minutes?”* or *“what is the probability that more than 10% of the population is infected within 2 weeks?”*

- Various efficient CTMC solution methods but importantly this area has driven the development of *statistical model checking*.
- Can be used to validate hypothesised models and can support *synthetic biology* (e.g. DNA computing)

Safety-Critical Systems

Used to rigorously quantify the likelihood with which high-risk failures or malfunctions could occur

- Assessment of safety integrity levels (SILs), which specify acceptable probabilities or rates of failure
- Modelled as DTMCs or CTMCs (for quantifying failure rates over time)
- Safety specifications typically amount to relatively simple PCTL or CSL queries

Early example: integration into the failure mode and effects analysis of an airbag system. Also used in:

- Railway infrastructure (using *Fault Tree Analysis*)
- Vehicle guidance systems
- Medical (pacemaker design) and aerospace domain (satellite systems and spacecraft design)

Autonomous Systems & Robotics

- Clear trend in recent years
- Robots operate in uncertain environments (also interaction with other agents) with imperfect sensors and actuators
- Guarantees of policies (safety and reliability) are crucial

Common use of LTL to capture complex temporal specifications like: *“maximise the probability of inspecting all three sensors, in any order, whilst avoiding X”*.

Often combined with **multi-objective model checking** $\Rightarrow$ allows generation of policies that trade-off between several, competing objectives (e.g., battery life vs. mission execution time) or analysis of the corresponding Pareto front.

- Use in AUVs, UAVs, self-driving cars, multi-robot systems
- Highlights challenges

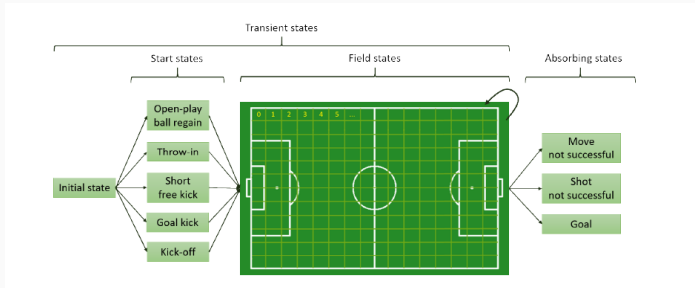
Some further application areas

- Human behaviour modeling
- Software architectures and self-adaptive systems
- Smart Grids (performance, resilience and security)
- Quantum computing (key distribution algorithms)
- Blockchain and cryptocurrencies
- Business process modeling

Some unconventional examples of applications: Sports tactics

Looking beyond the past: Analyzing the intrinsic playing style of soccer teams

Clijmans, J. et al., 2023

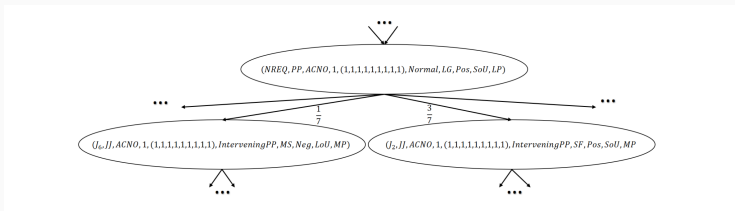


1. Learn an MDP model from event stream data
2. Apply probabilistic model checking with LTL to express outcomes of interest
3. Use the results to inform tactical decisions

Some unconventional examples of applications: Court interactions

Probabilistic model checking of temporal interaction dynamics in the supreme court

Das, S. and Sharma, A., 2023



1. Use transcripts for constructing a DTMRM
2. Formulate interesting queries over interactions by using PCTL
3. Verify queries by using a probabilistic model checker PRISM

Conclusion

- Commonalities between different problem domains
- Considerable breadth and diversity to the set of applications
- Ease of rigorous modelling for stochastic systems
- Logical formalism for expressing quantitative properties allows going beyond classical notion of correctness typically used in formal verification
- Continually evolving tools and possible future domains of application